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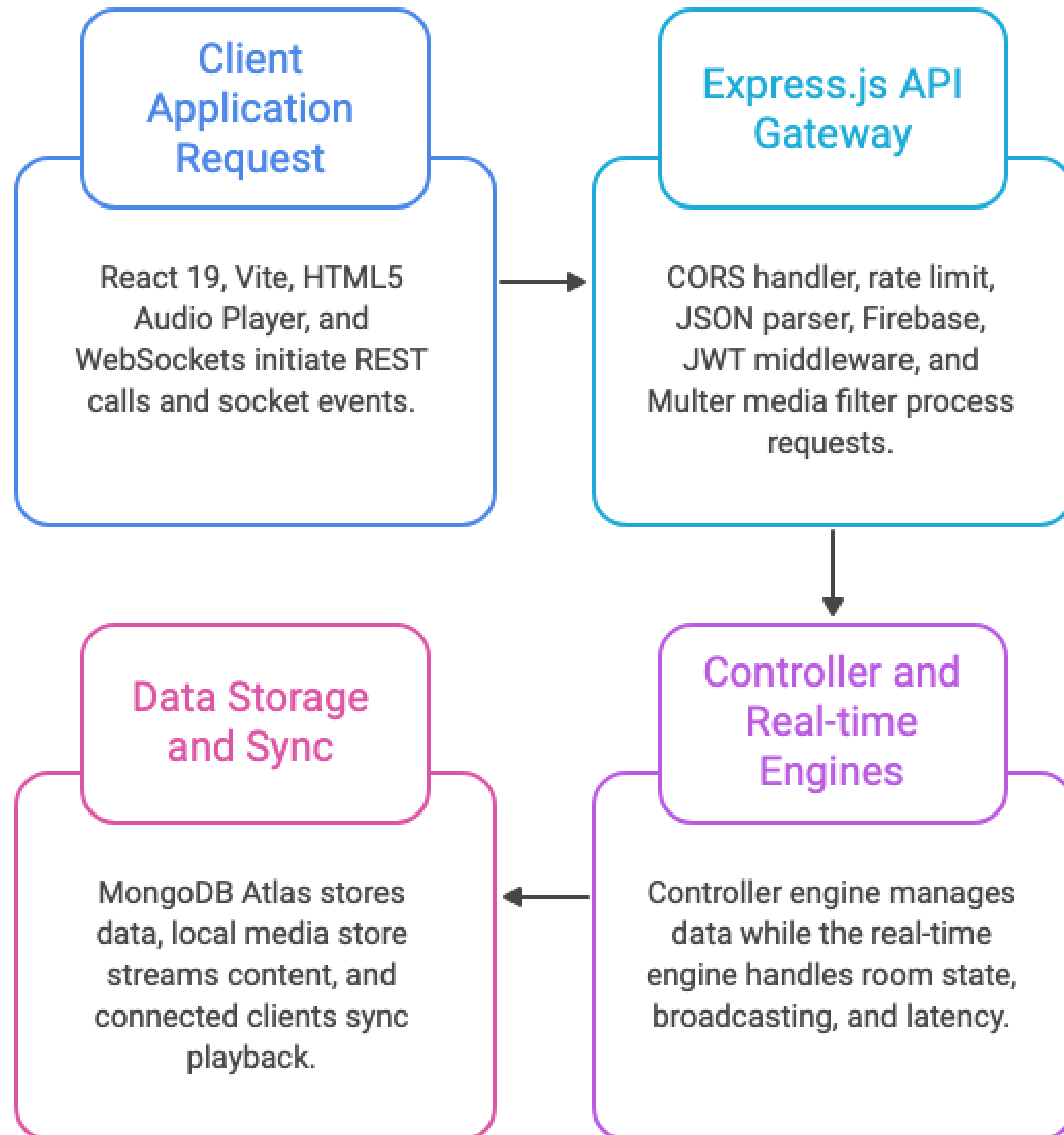
Backend, Node.js, Firebase, MongoDB

TuneWave : High-Fidelity Music Streaming Service

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High-Fidelity Music Streaming Service

Client Application Data Flow



- **3-Tier Decoupled Flow: Gateway Middleware → Modular Controllers → Persistence Layer.**
- **Dual-Protocol Delivery: REST for CRUD & metadata; WebSockets for zero-lag sync.**
- **High Efficiency: Native static streaming with HTTP Accept-Ranges byte-seeking.**

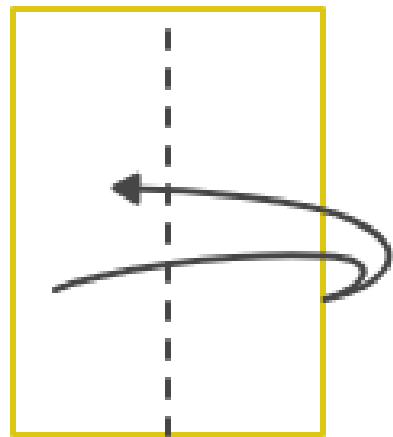


Dual Layer Authentication Flow

Security & Identity Pipeline
Firestore OAuth Identity + Stateless Backend JWT
Authorization

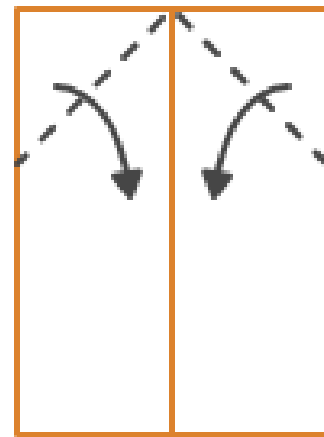
Security & Identity Pipeline
Firestore OAuth Identity + Stateless
Backend JWT Authorization

User Authentication Flow



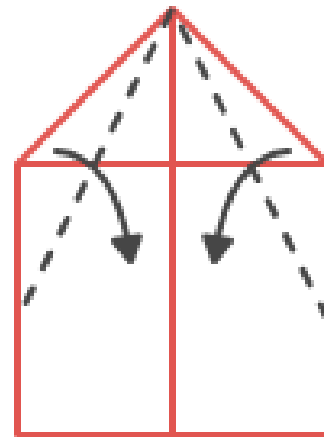
User Login

Submits
Google/Email
credentials



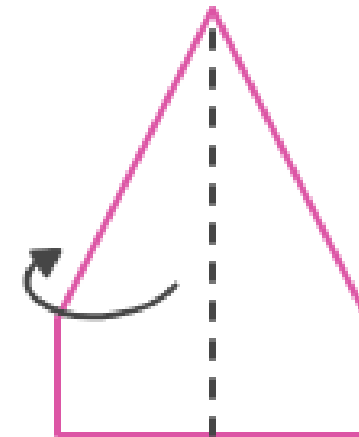
Firebase Auth

Generates signed
client ID token



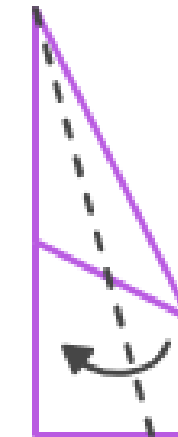
Express API

Verifies ID token
with Firebase Admin
SDK



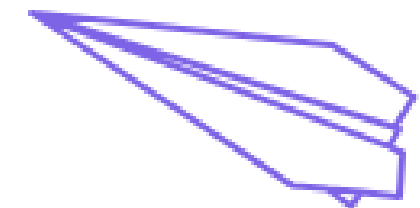
MongoDB Sync

Upserts user
document and
retrieves ID



Issue JWT

Generates signed
session token

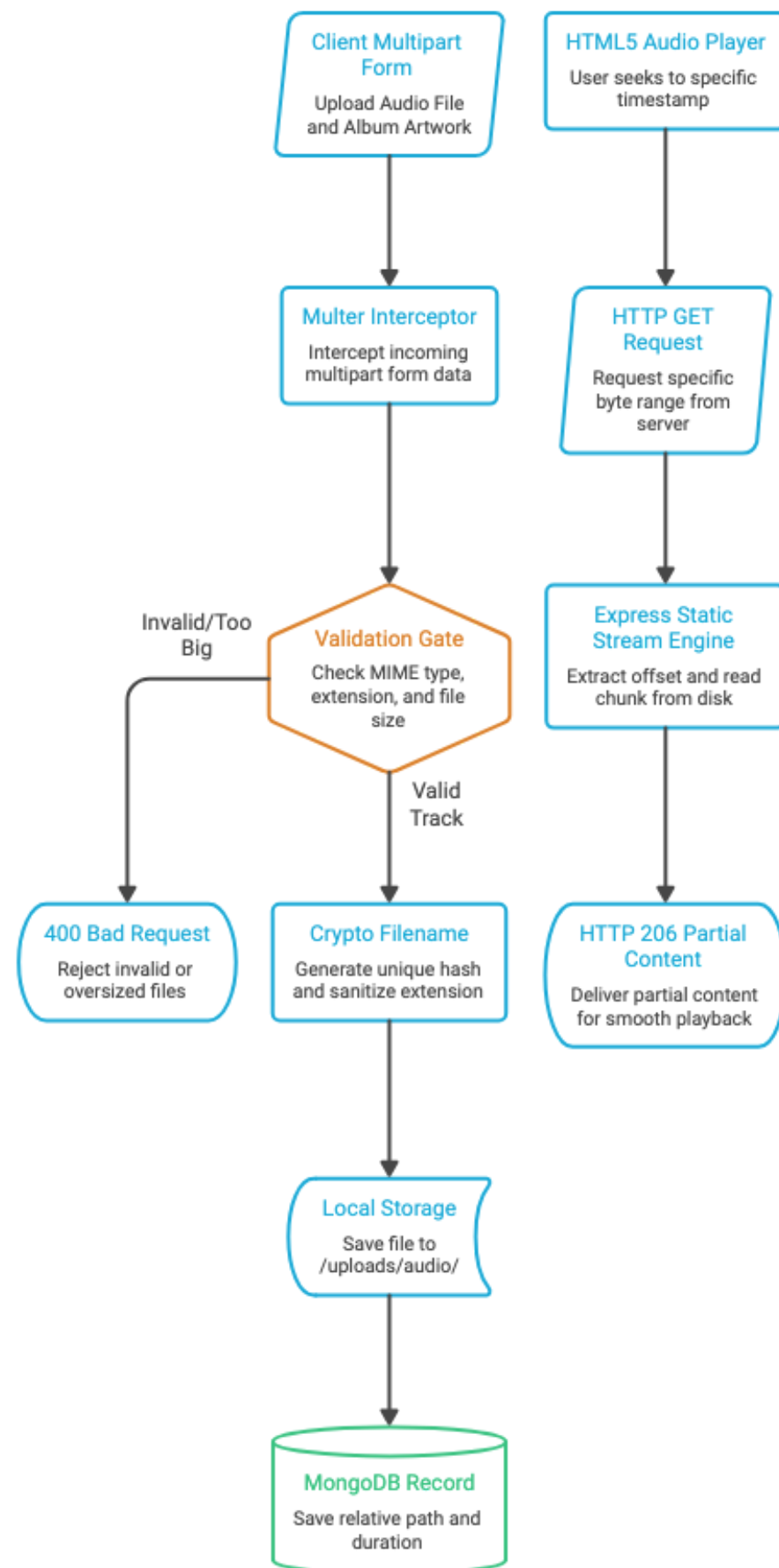


Protected Routes

Gatekeeper verifies
JWT for access

Audio Streaming & Upload Pipeline

Audio Processing and Streaming Pipeline



Audio Processing & Streaming Engine

MIME Sanitization Ingestion + HTTP 206 Partial Content Delivery

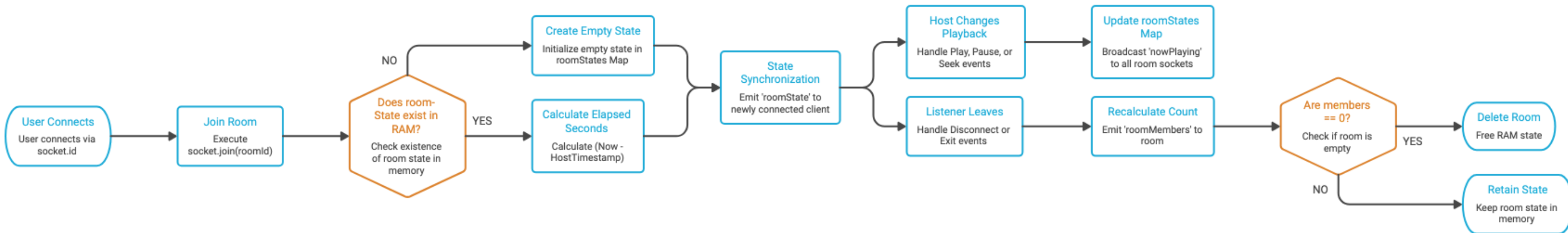
- **Instant Scrubbing:** Native HTTP 206 Range headers allow jumping to any timestamp instantly.
- **Zero Transcoding Delay:** Direct audio stream pipeline starts playback in under 150ms.
- **Storage Isolation:** Dedicated folder partitioning for tracks, album art, and artist portraits.



Real-Time Listening Room State Machine

Real-Time Group Sync Engine Socket.io Room Subscriptions with Math-Compensated Playback

Room State Management and Synchronization Flow



Sub-Millisecond Speed: $O(1)$ in-memory state lookup avoiding expensive database writes per tick.

Latency Math Compensation: New listeners sync to the exact playback second including network delay.

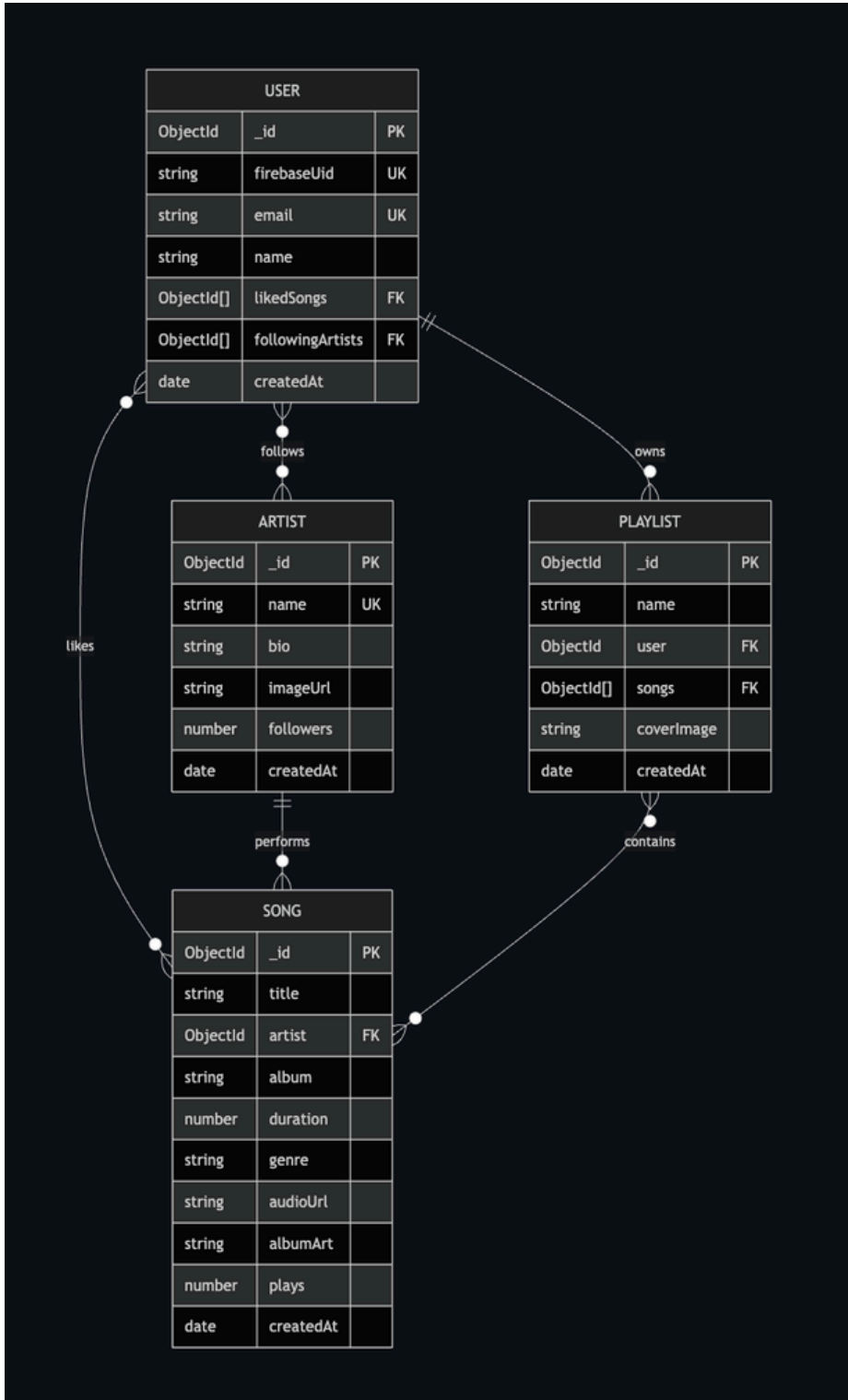
Auto Garbage Collection: Rooms automatically purge from memory when the last listener leaves.



Database Schema & API Ecosystem

Data Modeling & API Landscape

Relational Document Schema in MongoDB + Production Endpoints



CORE REST ROUTE MATRIX

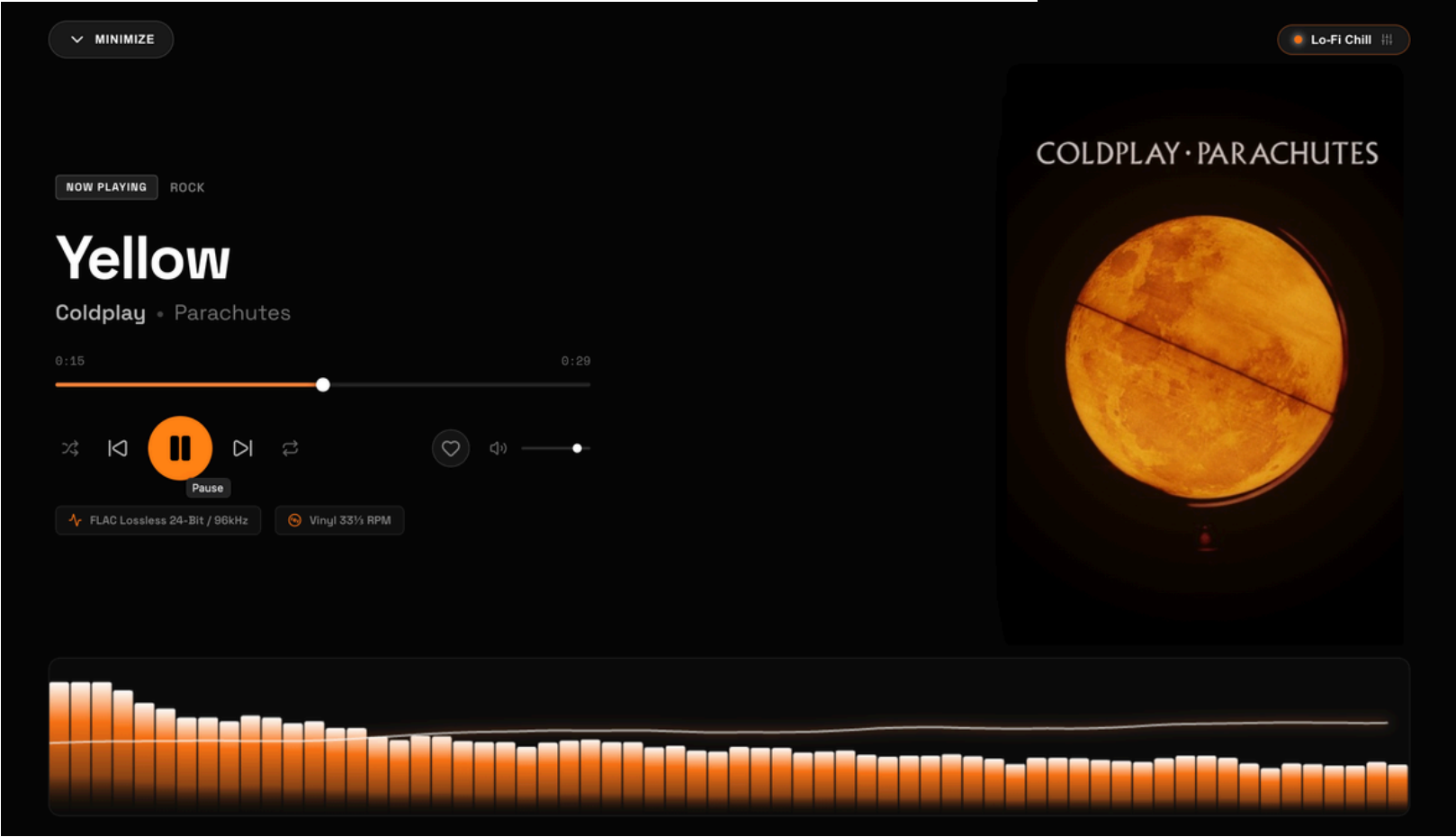
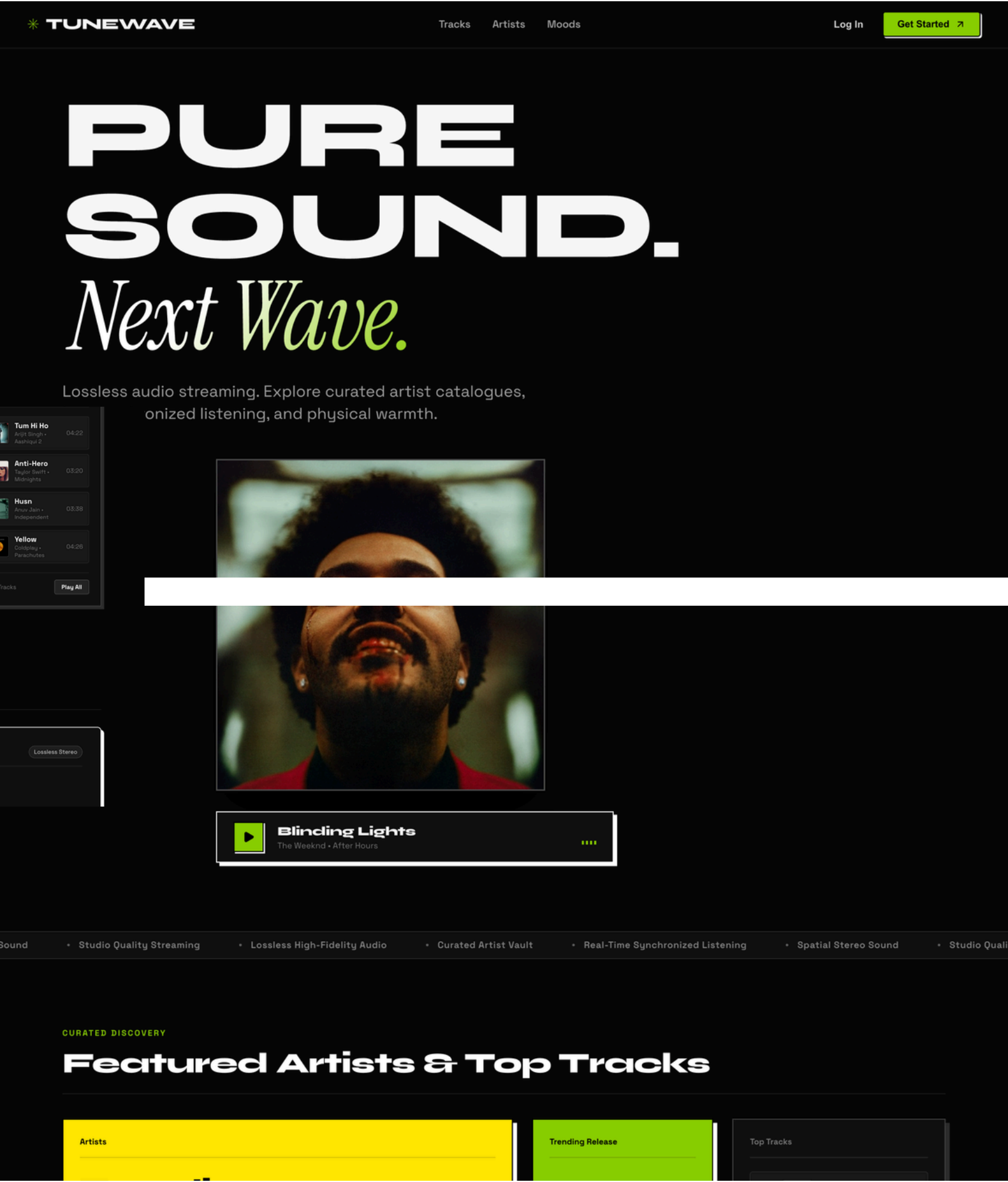
ROUTE PREFIX	/api/auth	/api/songs	/api/songs	/api/playlists	/api/playlists	/api/artists
METHODS & ENDPOINTS	POST /login, POST /register	GET /, GET /search, GET /:id	POST /upload	GET /my-playlists, POST /	PUT /:id, DELETE /:id	GET /, GET /:id, POST /follow
ACCESS CONTROL & FUNCTION	Public / Firebase Sync	Public Cached Catalog	Protected (Multer File System)	Protected JWT	Protected (Owner-Verified Only)	Public / Protected Social



TRY IT OUT !!!!

Vercel :[TuneWave-Frontend](#)

Render :[TuneWave-Backend](#)





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Thank You